Comparative Drug Discovery in Diabetic Nephropathy Using Bulk and Single-Cell Transcriptomics

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Abstract

Diabetic nephropathy (DN) remains a leading cause of end-stage renal disease worldwide, yet developing targeted therapies remains costly and challenging. This two-year study established a multi-scale computational framework to identify and prioritize small compounds capable of reversing DN gene signatures. In Phase 1, whole-tissue RNA-seq data from type 2 diabetic nephropathy (T2DN) renal samples were screened against drug-matching platforms, such as Playbook Workflow Builder, Connectivity Map, and iLINCS, to identify compounds that reverse diseaese-associated gene expression. Phase 2 expanded this methodology to single-nucleus resolution in early DN, evaluating how cell-specific signatures across podocytes, endothelial cells, and mesangial cells compared to whole-tissue bulk profiles. Computational matching prioritized molecules based on target reversal, statistical confidence, and mechanism of action. Cell-subtype analysis revealed distinct early disease mechanisms, including podocyte Senescence-Associated Secretory Phenotype (SASP) activation and endothelial hexose transport overload. Notably, podocyte-derived drug signatures were significantly enriched throughout Phase 2, accounting for 66.7% of compounds shared between single-cell and bulk datasets. Across both phases, the histone deacetylase (HDAC) inhibitor vorinostat consistently emerged as a top candidate, demonstrating capacity to reverse cell-specific signatures. These findings demonstrate that analyzing DN at single-cell resolution uncovers distinct drug targets hidden in whole-tissue data, while confirming that podocyte stress dominates early disease progression. Ultimately, this study offers an efficient, cost-effective approach for prioritizing lead therapeutics for future experimental validation.

Introduction

Diabetic nephropathy (DN) is one of the most common complications of diabetes mellitus and the leading cause of end-stage renal disease (ESRD) worldwide (Sinha & Pauniker, 2021). Often characterized by glomerular basement membrane thickening, mesangial matrix expansion, and progressive fibrosis, DN advances from hyperfiltration and microalbuminuria to a declining glomerular filtration rate, proteinuria, and ultimately ESRD (Bakris, 2011). With roughly 20–40% of individuals with diabetes developing DN globally, the condition places a heavy burden on healthcare systems and patient quality of life (Cao & Cooper, 2011).

Over the last two decades, theories have shown how chronic high blood sugar drives kidney damage through a mix of blood flow changes and metabolic stress, as depicted in Figure 1. These factors trigger intracellular signaling and transcription factors that release inflammatory markers and growth factors. Over time, this leads to extracellular matrix (ECM) accumulation, altered blood vessel permeability, and protein leakage in the urine (Forbes et al., 2007; McLennan et al., 2013).

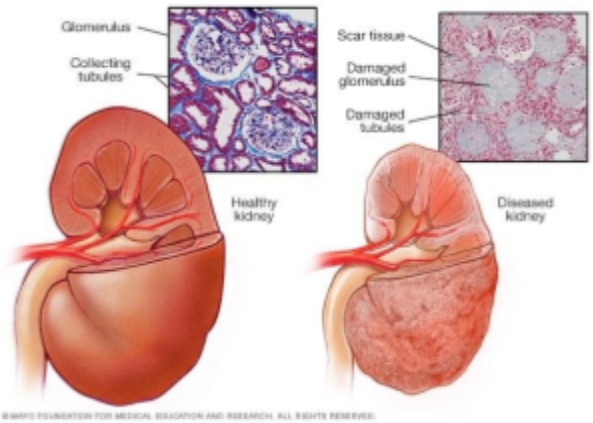


Figure 1. Comparison of a healthy kidney and a DN–affected kidney, highlighting structural damage, inflammation, and fibrosis (Mayo Clinic, 2023).

While traditional treatment relied on managing blood pressure and blood sugar, newer strategies focus on targeting glucose-dependent pathways directly to slow disease progression (McLennan et al., 2013). Preclinical and clinical studies show partial improvement by reducing oxidative stress, inflammation, and lipid buildup (Mima, 2013). However, stopping disease progression entirely remains a major challenge.

Complicating this issue is the high cost and difficulty of traditional drug discovery. Developing a single drug is a long, expensive process, with estimates ranging between $161 million and $4.54 billion per approved medication (Schlander et al., 2021). High failure rates often stem from a limited understanding of the exact biological mechanisms behind complex human diseases (Apic et al., 2005). Computational approaches offer a practical alternative to

lower development costs and identify existing compounds that can be repurposed for new treatments.

Phase 1: Whole-Tissue Drug Discovery in Type 2 DN (T2DN)

To address the need for new treatment options, the first phase of this study focused on type 2 diabetic nephropathy (T2DN), where prolonged high blood sugar damages the kidney's microvascular filtration barrier.

Problem Statement

- T2DN lacks effective disease-reversing therapies, and traditional whole-tissue RNA sequencing provides broad genetic profiles that fail to identify which small-molecule compounds can effectively reverse those disease signatures.

Objective

- The purpose of Phase 1 was to analyze whole-tissue RNA-seq data from T2DN renal samples to extract differentially expressed genes and screen computational drug databases for small molecules capable of reversing T2DN gene expression profiles.

Hypothesis

- If T2DN gene expression profiles are screened through computational drug-matching platforms, then specific small molecules can be identified that reverse the disease-associated gene signature and target DN pathways.

Phase 2: Single-Cell vs. Bulk Analysis in Early DN

Building upon the whole-tissue baseline established in Phase 1, Phase 2 expanded the computational framework through single-cell versus bulk-tissue comparisons, shifting focus to early DN to capture cell-type-specific drivers before permanent kidney damage occurs.

Problem Statement

- DN involves complex molecular changes across distinct kidney cell types, including podocytes, mesangial cells, and endothelial cells. Traditional bulk-tissue transcriptomic analyses—like those used in Phase 1—combine signals from all cell types together, potentially hiding cell-specific disease mechanisms and drug targets.

Objective

- The purpose of Phase 2 was to develop an efficient computational framework that integrates cell-type-specific and bulk-tissue transcriptomic data to:
    - Identify and compare potential therapeutic compounds for early DN.
    - Determine the biological pathways associated with these compounds to determine their therapeutic relevance.

Hypothesis

- If cell-type-specific transcriptional signatures reveal DN mechanisms that are obscured in bulk kidney tissue, then analyzing DN through individual kidney cell types will identify distinct therapeutic compounds and biological processes compared with bulk-tissue analysis.

This two-year study demonstrates the value of a systematic computational approach for finding and prioritizing candidate drugs, saving time and resources. Connecting candidate compounds to their biological pathways offers clearer insights into how these drugs work, helping guide future experimental testing. Additionally, because this framework uses public datasets and standard pathway tools, it can easily be adapted to study other complex diseases.

Materials & Methods

Over the course of 69 weeks, during the past two summers, I was responsible for developing the methodology framework, collecting data, and comparing results from various DN-related RNA-sequencing datasets. My mentor, Dr. Charles Mobbs, only provided general suggestions on computational platforms I should use, such as CMap and PWB.

Phase 1: Whole-Tissue Transcriptomic Screening in Type 2 Diabetic Nephropathy

Focusing initially on drug identification for T2DN, Phase 1 involved two computational approaches using whole-tissue RNA-seq data from the study *"Identification of Smad3-related*

transcriptomes in type-2 diabetic nephropathy by whole transcriptome RNA sequencing" (Zhou et al., 2020), which compared T2DN kidney samples against non-diabetic controls.

In the first approach, the top 150 upregulated genes were uploaded to the LINCS L1000 Signature Search (L2S2) within Playbook Workflow Builder (PWB), a web-based bioinformatics platform (Clarke et al., 2025). The top fifty small molecules with the highest odds ratio and strongest inverse transcriptional signatures to T2DN were retained for resource efficiency.

For further validation, the same top 150 upregulated genes inserted in L2S2 and an additional 150 downregulated genes from the original dataset were analyzed using the drug repurposing tool CLUE Platform (CMap) (Samart et al., 2021). Compounds with strongly negative tau scores and transcriptional impact scores (TIS) $\geq 90th$ percentiles were prioritized and cross-referencing with L2S2 to yield the three top-ranking molecules.

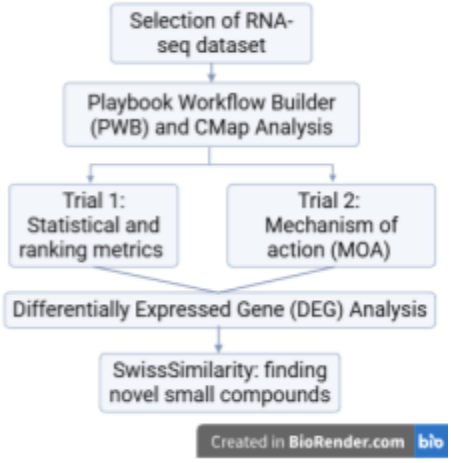


Figure 2. Workflow for Phase 1, including DEG extraction, transcriptomic screening, and compound selection *(Figure generated by student researcher using BioRender.com, 2025).*

The second computational approach began with the literature-backed identification of T2DN-related mechanisms of action (MOAs), including inflammation, oxidative stress, fibrosis, and glucose metabolism (Młynarska et al., 2025). Then, using L2S2 and CMap, compounds were prioritized by their MOAs, followed by the same statistical filters as the first trial (Figure 2).

The gene expression from the compounds identified across both approaches were compared against other diabetic complications, including diabetic retinopathy (DR) and diabetic neuropathy (DNEURO) to understand particular genes that were targeted (Table 1). Overlapping genes, or genes expressed in both the compound and disease, were identified in Google Sheets using the function

$$= COUNTIF(\$A\$1{:}\$I\$202,\ A1) > 1.$$

Oppositional gene changes (e.g. downregulated in disease but upregulated by a compound) underwent literature review via PubMed to confirm biological relevance.

Table 1

Gene Expression Omnibus (GEO) Datasets Used for Differential Expression Analysis

GEO Accession	Disease Studied
GSE30528	Diabetic Nephropathy
GSE53257	Diabetic Retinopathy
GSE185011	Diabetic Neuropathy

Note. Table generated by student researcher using Google Slides, 2025.

Beyond confirmation, the objective of this phase was to find novel small compounds of therapeutic potential. Using SwissSimilarity, a ligand-based virtual screening platform, compounds of similar structure to the top L2S2 and CMap were identified. Simplified Molecular Input Line Entry System (SMILES) representations of each compound were screened and measured using Tanimoto coefficients, with scores ≥ 0.8 considered high similarity. Further literature study was completed on these results.

Phase 2: Cell-Specific and Whole-Tissue Transcriptomics in Early Diabetic Nephropathy

Building on Phase 1, Phase 2 expanded the drug-identification framework to single-cell analysis in early DN. This modification was intended to capture cell-type-specific drivers of T2DN prior to any irreversible structural damage and evaluate how subtype signals interact in whole-tissue profiles.

Analysis focused on three glomerular cell types involved in filtration and blood flow: podocytes, endothelial cells, and mesangial cells. To find DEGs within each subtype, a public scRNA-seq dataset from *"The single cell transcriptomic landscape of early human diabetic nephropathy,"* which contained 23,980 transcriptomes across three control and three early DN human kidney samples, was used (Wilson et al., 2019). Cell-specific DEGs were organized in Google Sheets and filtered based on statistical significance using

$$= AND(ABS(C3) > 0.25,\ F2 < 0.05),$$

indicating an absolute average log fold change $(avg_logFC) \geq 0.25$ in Column C and an adjusted p-value ≤ 0.05 in Column F. Up- and down-regulated genes were categorized using

$$= IF(C1 > 0.25,\ "Up,"\ IF(C1 <- 0.25,\ "Down",""))).$$

Cross-dataset overlaps were flagged using

$$= COUNTIF(\$A\$1{:}\$I\$202,\ A1) > 1.$$

Genes sharing expression directions across multiple subtypes were identified for literature validation to confirm functional relevance.

Pathways in Specialized Cells

Significant dysregulated genes for each cell type were queried in the biological pathway database Reactome and g:Profiler using the Gene List Functional Profiling tool (*Homo sapiens* background, incorporating Reactome and KEGG pathways) (Raudvere et al., 2019). Results were prioritized by frequency, lowest p-values (< 0.05), and lowest FDR scores (< 0.05) and PubMed searches using pathway terms alongside "fibrosis," "DN," and "CKD" helped further evaluate pathways shared across subtypes. Non-overlapping pathways were analyzed similarly to capture unique cell functions. Finally, a Shared vs. Cell-Specific Biological Pathways heatmap was created in Excel using $-\ log_{10}(p\ -\ value)$ scores with a blue-to-red color gradient.

Drug Identification

Candidate therapeutics were identified using iLINCS, with up- and downregulated gene sets serving as input profiles. Output data included perturbagen names, p-values, and concordance scores (scaled $-\ 1$ to $+\ 1$), with negative scores approaching $-\ 1$ prioritized for signature reversal. Two selection strategies were applied:

1. Concordance Approach: Selected compounds with concordance scores closest to $-\ 1$.
2. Gene-overlap Approach: Selected compounds targeting the highest number of overlapping disease genes. Therefore, a compound reversing 10 podocyte DN genes was prioritized over a compound reversing only 3 genes).

Differentially Expressed Genes (DEGs) and Pathway Reversal in Compounds

To evaluate which compounds most successfully reversed disease-associated signatures, iLINCS datasets containing the compound-specific DEGs in kidney tissue were used (Table 2). Genes exhibiting statistically significant ($p < 0.05$, $|logFC|\ > 1|$) and opposite expressions relative to the initial DN dataset were identified in Google Sheets. And, using regular expression matching

$$= REGEXMATCH(A1,\ "\text{^}(?\,i)(GENE1|GENE2|\ldots)")$$

upregulated DN genes were cross-referenced against the downregulated expression profiles of each compound (and vice versa).

Compounds were then prioritized based on gene-overlap, and given the size of the initial disease gene set, a minimum threshold of $n = 3$ reversed genes was applied. Therefore, compounds failing to meet this threshold were excluded from future study.

For qualifying compounds, the up- and downregulated gene sets were separately analyzed in Reactome to identify drug-affected pathways. Pathways were filtered using a threshold of $p < 0.05$ and structured into two sets:

- Sheet A: Unique upregulated pathways across compounds.
- Sheet B: Unique downregulated pathways across compounds.

Table 2.
Library of Integrated Network-Based Cellular Signatures (LINCS) Datasets Used for Compound-Specific Differential Expression Analysis

Compound	LINCS Dataset (Kidney Epithelial Tissue)
Troglitazone	LINCS66844
Raloxifene	LINCS17483
Valproic Acid	LINCS15322
Calcitriol	LINCS212354
Imatinib	LINCS16560
Vorinostat	LINCS15424
Niclosamide	LINCS15312
Nicardipine	LINCS15176
Cyclosporine	LINCS207922

Note. Table generated by student researcher using Google Docs, 2026.

Pathways from Sheets A and B were cross-referenced against the list of DN disease pathways to flag opposite expression states (i.e. upregulated in disease and downregulated by drug). Due to nomenclature variations across databases, pathways were mapped based on shared biological themes (e.g. senescence regulation, proliferation signaling, extracellular matrix remodeling).

Bulk Tissue Transcriptomic Validation

Upon the completion of single-cell analysis, a nearly identical framework was applied to bulk RNA-seq data using dataset GSE142025 (*"Integrated analysis of single-cell RNA-seq and bulk RNA-seq combined with multiple machine learning identified a novel immune signature in diabetic nephropathy"*) (Peng et al., 2023). Within the toolset GEO2R, DEGs were identified across 9 control and 6 early DN samples. To manage the large dataset, which contained over 26,00 preliminary DEGs, stricter statistical thresholds were applied: adjusted $p < 0.001$ and $|log_2 FC| > 1.5$. Filtered DEGs were analyzed in Reactome for pathway enrichment and queried in iLINCS to extract the top bulk-derived therapeutic compounds.

Single-Cell vs. Bulk Tissue Comparison

Compounds accumulated from both approaches were categorized into three sets: Shared (across single-cell and bulk-tissue), Unique Single-Cell, and Unique Bulk. All compounds found from single-cell analysis were mapped to their originating cell subtypes (podocyte, endothelial, or mesangial). To evaluate cell-type representation in each set, Fisher's exact test assessed the statistical significance of any "overexpression" of specific cell types or compounds. An odds ratio was also calculated in Python to quantify the likelihood of any cell-specific-derived compounds appearing exclusively in the Unique Single-Cell list.

Finally, all DN-relevant pathways accumulated in Phase 2 were cross-compared across single-cell disease, bulk disease, single-cell drugs, and bulk drugs to identify inverse expression patterns. In other words, all pathways were categorized by driver profile (endothelial-, mesangial-, podocyte-, or bulk-driven) alongside logFC values to determine which single-cell and bulk-tissue drug signatures—and thus which drugs—most strongly opposed disease-expression (Figure 3).

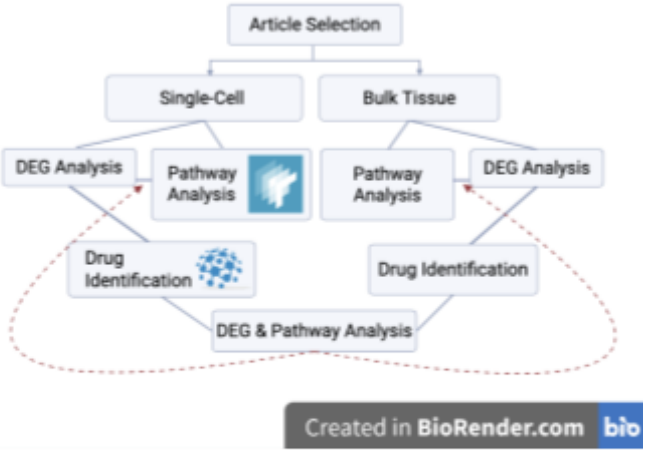


Figure 3. Workflow for Phase 2, including single-cell and bulk-tissue DEG extraction, pathway analysis, and drug identification *(Figure generated by student researcher using BioRender.com, 2026).*

Results

Phase 1: Whole-Tissue Transcriptomic Screening in Type 2 Diabetic Nephropathy

Compound Identification Using Two Computational Approaches

In the first approach to T2DN drug discovery, compounds were prioritized by statistical performance and then evaluated through literature review and known MOAs. Two compounds–pyrvinium-pamoate (PP) and camptothecin (CPT)–consistently ranked as strong candidates. CMap analysis showed highly negative tau scores for both (PP: -99.93; CPT: -99.72),

showing strong reversal of T2DN-associated transcriptional signatures. Additionally, unlike the typical odds ratios between 2 and 5, which was observed across most screened compounds, PP and CPT both exhibited odds ratios of 23.8 in L2S2. Their transcriptional impact scores (PP: 94th percentile; CPT: 99th percentile) further supported their relevance (Table 3).

Table 3

Tau Scores, Odds Ratios, and Transcriptional Impact Scores for Prioritized Compounds in Trial 1

Compound	Tau Score	Odds Ratio	Transcriptional Impact
Pyrvinium Pamoate (PP)	-99.93	23.8	0.67 (94th percentile)
Camptothecin (CPT)	-99.72	23.8	0.94 (99th percentile)

Note. Table generated by student researcher using Google Docs, 2025.

The second approach began with prioritizing compounds with MOAs relevant to T2DN–specifically histone deacetylase (HDAC) inhibition, farnesyltransferase inhibition, and suppression of NF-κB-mediated inflammation—before statistical filtering to emphasize biological significance over mere numerical confidence in the selection of candidate therapeutics.

Three compounds emerged: vorinostat (HDAC inhibitor), manumycin A (farnesyltranferase and NF-κB inhibitor), and alvespimycin (HSP90 inhibitor). CMap and L2S2 results supported their potential (Table 4).

Preclinical evidence, found through literature review, demonstrated that all five compounds—PP, CPT, vorinostat, manumycin A, and alvespimycin—demonstrated extensive metabolic relevance, as noted in the Discussion.

Table 4

Tau Scores, Odds Ratios, and Transcriptional Impact Scores for Prioritized Compounds in Trial 2

Compound	Tau Score	Odds Ratio	Transcriptional Impact
Vorinostat	-98.98	19.8	0.79 (96th percentile)
Manumycin A	-99.58	22.7	0.73 (95th percentile)
Alvespimycin	-97.92	15.1	0.65 (94th percentile)

Note. Table generated by student researcher using Google Docs, 2025.

DEG Analysis and SwissSimilarity

DEG comparisons across multiple GEO datasets further supported the relevance of these compounds. Notably, genes HSP90AA1 and HSP90B1, both related to the reduction of ER stress and apoptosis, were upregulated by alvespimycin and manumycin A and downregulated in T2DN datasets (Hoter et al., 2018). Similarly, while IFITM3 was elevated in T2DN and diabetic neuropathy, it was downregulated by vorinostat, supporting the therapeutic relevance of HDAC inhibition (Table 5).

Following the identification of compounds with therapeutic potential, SwissSimilarity identified structurally similar compounds with similarity scores above 0.8, including the HDAC inhibitor pyroxamide (similarity: 0.984), mofebutazone (similarity: 0.842), and bucetin (similarity: 0.828).

Table 5

Comparison of Compound-Regulated Genes from Trial 2 with DEGs in Diabetic Complication Datasets

Gene	Upregulated	Downregulated
HSPA5	Alvespimycin	DN, DR
PROM1	Manumycin A, DR, DNEURO	N/A
HSP90AA1	Alvespimycin	DN, DR
PINK1	Vorinostat, DR, DNEURO	N/A
NPTN	Manumycin A	DN, DR
IFITM3	DN, DNEURO	VOR
HSP90B1	Manumycin A	DN, DR
RPL23	Vorinostat	DR, DNEURO

Note. Table generated by student researcher using Google Docs, 2025.

Phase 2: Cell-Specific and Whole-Tissue Transcriptomics in Early Diabetic Nephropathy

Subtype-Specific Differentially Expressed Genes (DEGs)

Initial DEG screening across the three specialized cell subtypes identified 281 total significant genes, with 21 genes (8.93%) identified in more than one subtype. A further breakdown of cell-specific composition can be found in Table 6.

Four DEGs maintained identical expression directions across all three subtypes. *HDAC9*, *NR4A1*, and *UTY* were universally upregulated, whereas *DHRS3* was universally downregulated. Conversely, two genes exhibited opposing cell-type expression: *TCF4* was upregulated in mesangial cells but downregulated in endothelial cells, and *ADAMTS6* was downregulated in mesangial cells and upregulated in endothelial cells.

Disrupted Pathways in Specialized Cells

Querying DEGs in Reactome gave distinct pathway profiles across each cell subtype. Within podocytes, only 7 pathways met the significance threshold ($p < 0.05$) out of over 700 total outputs. Meanwhile, endothelial DEGs generated a substantially larger yield with over 1,000 outputs, producing 34 upregulated and 25 significant downregulated pathways with lower FDR values

Table 6

Number of Upregulated and Downregulated Genes Across Podocytes, Endothelial Cells, and Mesangial Cells

Cell Subtype	Upregulated Gene Count	Downregulated Gene Count
Podocyte	35	37
Endothelial	98	47
Mesangial	45	19

Note. Table generated by student researcher using Google Docs, 2026.

and higher entity counts. Finally, mesangial DEGs yielded 30 upregulated and 23 downregulated significant pathways.

g:Profiler enrichment queries across up- and downregulated gene sets yielded relatively low pathway counts (podocytes: $n = 12$; endothelial: $n = 157$; mesangial: $n = 30$) and low biological specificity for renal pathology. For instance, top podocyte pathways included broad developmental classifications, rather than unique disease-associated pathways, such as *"multicellular organismal process"* and *"multicellular organism development."* Due to this lack of specificity for DN, g:Profiler outputs were excluded from further research analysis.

Within the 126 significant Reactome pathways identified from across all three subtypes, 14 were found to be shared between multiple cell types. Endothelial and mesangial cells accounted for 12 of these shared pathways, whereas podocytes overlapped with other subtypes in only two pathways: *Senescence-Associated Secretory Phenotype (SASP)* and the *Terminal pathway of complement*. The relative significance, expressed by the $-\log_{10} FDR$ score, of each shared pathway is summarized in Figure 4.

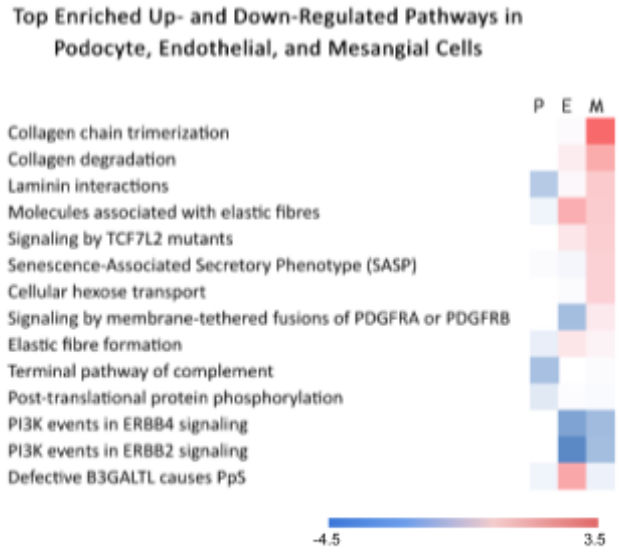


Figure 4. Heat map visualization of the pathway expression across podocyte (P), endothelial (E), and mesangial (M) subtypes. *(Figure generated by student researcher using Excel, 2026).*

Transcriptomic Perturbation Screening & Drug Identification

Cell-specific DEG sets were queried in iLINCS to identify small-molecule perturbagens capable of disease-associated expression reversal. Inputting podocyte DEGs yielded 10,393 LINCS profiles ($n = 4$ overlapping genes) and 31 CMap profiles ($n = 45$ overlapping genes). Endothelial DEGs yielded 14,337 LINCS and 51 CMap profiles ($n = 108$ overlapping genes). Mesangial DEGs yielded 8,186 LINCS ($n = 5$ overlapping genes) and 26 CMap profiles ($n = 48$ overlapping genes).

The top 25 candidate compounds per cell subtype were chosen from the LINCS outputs based on their statistical relevance to DN-specific MOAs, established earlier in the paper. CMap

profile results were incorporated by including compounds with the same criteria for DN-relevant MOAs, adding 5 podocytes, 16 endothelial, and 20 mesangial compounds to their respective subtypes. Combining the LINCS and CMap lists amounted to a total of 118 compounds with therapeutic potential. Six of these compounds appeared at least twice and prioritized: troglitazone (endothelial-derived), valproic acid (mesangial-derived), raloxifene (podocyte-derived), calcitriol (endothelial-derived), imatinib (mesangial-derived), and vorinostat (podocyte-derived).

Table 7

Number of Significant Differentially Expressed Genes for Compounds Derived from Single-Cell RNA-Seq Data, Categorized by Expression Direction

Compound	Upregulated Gene Count	Downregulated Gene Count
Troglitazone	151	117
Valproic Acid	120	74
Raloxifene	61	70
Calcitriol	31	44
Imatinib	87	135
Vorinostat	282	269

Note. Table generated by student researcher using Google Docs, 2026.

Dysregulated Genes and Unique Pathway Reversal

Significant DEG profiles for each prioritized compound were larger than the original DN disease DEG sets (Table 7). And when cross-referencing the DEGs of the compounds against those of DN, 12 key genes were repeated with opposite expression (Table 8). For example, *CDK6* was consistently downregulated across three compounds (raloxifene, imatinib, and vorinostat), whereas *TIPARP* was downregulated exclusively by vorinostat. Across these 15 inverse expression instances, podocyte-derived compounds accounted for the majority of target reversals (60%, 9/15), divided between vorinostat (40%, 6/15) and raloxifene (20%, 3/15). Troglitazone (endothelial-derived) accounted for 20% (3/15), and imatinib (mesangial-derived) accounted for 20% (3/15).

Table 8

Comparison of Disease Gene Expression Profiles in Early Diabetic Nephropathy with Compound-Regulated Signatures

Gene	DN Expression	Drug Expression	Compound Source
PLSCR1	UP	DOWN	Troglitazone
CDK6	UP	DOWN	Raloxifene, Imatinib, Vorinostat
PRCP	UP	DOWN	Valproic Acid
COL4A1	UP	DOWN	Vorinostat
MEF2C	UP	DOWN	Valproic Acid
BHLHE340	UP	DOWN	Vorinostat
INPP4B	UP	DOWN	Vorinostat
MTHFD2	DOWN	UP	Troglitazone
TSC22D3	DOWN	UP	Raloxifene, Troglitazone
PAX8	DOWN	UP	Raloxifene
DDIT4	DOWN	UP	Vorinostat
TIPARP	DOWN	UP	Vorinostat

Note. *Table generated by student researcher using Google Docs, 2026.*

Pathway Analysis

In regards to pathway analysis, the DEGs of six single-cell-derived compounds yielded 1,070 total Reactome pathways (630 upregulated, 440 downregulated), with further breakdown in Table 9. Among the pathways shared by at least two compounds, 229 were upregulated and 352 were downregulated. Excluding pathways exhibiting bidirectional expression yielded 89 strictly upregulated and 105 strictly downregulated shared pathways.

Table 9

Number of Significant Differentially Expressed Pathways for Compounds Derived from Single-Cell RNA-Seq Data, Categorized by Expression Direction

Compound	Upregulated Pathway Count	Downregulated Pathway Count
Troglitazone	206	117
Valproic Acid	52	74
Raloxifene	83	70
Calcitriol	18	44
Imatinib	52	135
Vorinostat	219	80

Note. Table generated by student researcher using Google Docs, 2026.

Compound-unique responses exhibited notable reversals of cell-type-specific stress markers. For instance, in podocytes, the pathway *Cellular Senescence*, which was downregulated in DN, was upregulated by the compound vorinostat. By contrast, mesangial-specific pathways lacked direct, single-gene opposite-expression pathways shared consistently across multiple compounds.

Disease-Upregulated vs. Drug-Downregulated Reversals

The strongest inverse alignment between disease drivers and drug suppression occurred within extracellular matrix (ECM) and fibrotic pathways. Disease-upregulated pathways—including *collagen chain trimerization*, *laminin interactions*, *elastic fibre formation*, *crosslinking of collagen fibrils*, and *assembly of collagen fibrils and other multimeric structures*—were counteracted by drug-downregulated pathways including *collagen formation*, *assembly of collagen fibrils*, *ECM proteoglycans*, and *non-integrin membrane-ECM interactions*.

Disease-Downregulated vs. Drug-Upregulated Reversals

This pattern was persistent, particularly within pathways involving mitogen-activated protein kinases (MAPK), protein enzymes responsible for ordering cellular growth, division, or death. For instance, disease-downregulated pathways—*regulation of MAPK cascade*, *positive regulation of MAPK cascade*, and general *MAPK cascade*—were counteracted by

drug-upregulated pathways including *MAP kinase activation, MAPK targets/Nuclear events mediated by MAP kinases, SHC-mediated cascade:FGFR2, FRS-mediated FGFR2 signaling, and Signaling by PDGF.*

Bulk Tissue Screening and Single-Cell Cross-Comparison

DEG analysis of bulk RNA-sequencing dataset GSE142025 (adjusted $p < 0.001$, $|log_2FC| > 1.5$) identified 245 DEGs (73 upregulated, 171 downregulated). Pathway enrichment in Reactome yielded 166 upregulated and 634 downregulated pathways.

Integrative screening across iLINCS and CMap identified 724 unique bulk-tissue candidate compounds. Cross-referencing these against the 112 unique drugs derived from single-cell analysis identified 15 shared compounds. Within this group, 6 compounds overlapped with the statistically prioritized single-cell candidate set, 7 were unique to single-cell analysis, and 9 were unique to bulk-tissue analysis (Table 10).

Podocyte-derived candidates showed a distinct distribution in the shared set: 4 out of 6 (66.7%) were podocyte-derived, whereas podocyte-derived drugs were absent (0%) in both the unique single-cell and unique bulk candidate sets. Among all initial podocyte-derived drugs ($n = 30$), 13.33% ($4/30$) were shared with bulk-tissue data, compared to only 2.44% ($2/82$) for non-podocyte (endothelial and mesangial) drugs. Fisher's exact test confirmed this enrichment was statistically significant ($p = 0.04317$), and

Table 10

Compound Replication Status Across Studies and Cell Subtypes

Compound	Status	Cell Subtype	Compound	Status
Valproic Acid	Shared	Podocyte	Thioridazine	Unique (Bulk-Tissue)
Raloxifene	Shared	Podocyte	Azacyclonol	Unique (Bulk-Tissue)
Vorinostat	Shared	Podocyte	Niclosamide	Unique (Bulk-Tissue)
Wortmannin	Shared	Podocyte	Perphenazine	Unique (Bulk-Tissue)
Brefeldin A	Shared	Endothelial	Nicardipine	Unique (Bulk-Tissue)
Imatinib	Shared	Mesangial	Ipratropium	Unique (Bulk-Tissue)
Olaparib	Shared	Podocyte	Penfluridol	Unique (Bulk-Tissue)
Calcitriol	Unique (Single-Cell)	Podocyte	Cyclosporine	Unique (Bulk-Tissue)
Troglitazone	Unique (Single-Cell)	Podocyte	Nefazodone	Unique (Bulk-Tissue)
Pirinixic Acid	Unique (Single-Cell)	Endothelial		
Tyrphostin AG-1478	Unique (Single-Cell)	Endothelial		
Cyclosporine	Unique (Single-Cell)	Endothelial		

Note. Table generated by student researcher using Google Docs, 2026.

the calculated odds ratio ($OR = 6.15$) indicates that podocyte-derived drugs were approximately 6.15 times more likely to be shared with bulk-tissue profiles than non-podocyte-derived drugs.

Discussion

Phase 1 Reversal of T2DN

The first phase of this study showed that bulk-tissue transcriptomic profiling can successfully identify small-molecule therapeutics capable of reversing whole-tissue T2DN signatures. Initial screening identified compounds that target core pathological drivers of T2DN. For instance, PP acts as an anthelmintic agent that activates AMP-activated protein kinase (AMPK) and inhibits β-catenin, effectively suppressing gluconeogenic and lipogenic gene expression to reduce metabolic stress (Ishii et al., 2012; Schultz et al., 2021). Similarly, camptothecin and manumycin A modulate nuclear factor kappa B (NF-κB) and Ras-mediated proliferative pathways to limit pro-inflammatory cytokine expression (Huang et al., 2000; Cecrdlova et al., 2016). The virtual ligand screening conducted via SwissSimilarity identified structural analogs like quercetin, which exhibits established renal-protective effects by suppressing oxidative stress and cellular inflammation (Chen et al., 2022).

Phase 2 Reversal of Early DN

Across specialized glomerular subtypes, early DN pathology was marked by distinct transcriptional disruptions. Reactome showed that endothelial and mesangial cells elevated cellular hexose transport (GLUT1 overexpression) and altered laminin-collagen trimerization, which, according to published research, drives hyper-glycolysis and matrix accumulation (Lu et al., 2021). Simultaneously, podocytes and mesangial cells increased SASP activation, which has been found to sustain local inflammatory and TGF-β exposure (Hejazian et al., 2024). Together, these disruptions directly drive the metabolic stress, matrix expansion, and chronic inflammation that define early DN progression.

More importantly, the candidate small molecules identified across single-cell subtypes actively reversed these pathologies. HDAC inhibitors (such as podocyte-derived vorinostat)

suppressed pro-fibrotic gene expression and ECM deposition (Advani et al., 2011). Vitamin D receptor agonists like calcitriol attenuated chronic high-glucose-induced NF-κB signaling in endothelial cells, reducing downstream inflammatory mediators including TNF-α, IL-6, and ICAM-1 (Chagas et al., 2012).

Podocyte Dominance in Early Disease

A central finding of this study was the significant enrichment of podocyte-derived drug signatures among compounds found across both single-cell and bulk-tissue profiles. Podocyte-derived signatures constituted 66.7% (4/6) of the candidates shared between single-cell and bulk datasets, whereas podocyte-derived candidates were entirely absent (0%) among unique single-cell and unique bulk candidate sets. Fisher's exact test confirmed that this cross-scale overlap was statistically significant ($p = 0.04317$), with an odds ratio of 6.15 indicating that podocyte-reversing drug candidates were over six times more likely to remain detectable at the whole-tissue bulk level than candidates derived from endothelial or mesangial subtypes.

This representation directly reflects fundamental renal pathology. Podocytes are terminally differentiated, post-mitotic cells that form the primary filtration barrier of the glomerulus, so unlike glomerular endothelial and mesangial cells—which retain proliferative capacity and plasticity in response to metabolic stress—podocytes cannot readily undergo cell division or replenish lost population numbers (Barutta et al., 2022). Consequently, podocyte stress and subsequent podocytopenia, or an abnormal decrease in the number of kidney podocytes, represent an irreversible tipping point in diabetic nephropathy, driving severe proteinuria, glomerulosclerosis, and progressive decline in renal function (Barutta et al., 2022).

Thus, intervening pharmacologically at this early stage of DN to maintain podocyte adhesion and prevent cell loss indicates a promising strategy for slowing DN progression before permanent structural scarring occurs. The ability of this framework to analyze both shared multi-scale signals and tissue-specific profiles validates its capacity to prioritize reproducible therapeutic candidates while preserving biological resolution.

Therapeutic Prioritization and Lead Candidate Profile: Vorinostat

Among the candidate small molecules identified across analytical levels—including raloxifene, vorinostat, olaparib, wortmannin, brefeldin A, and imatinib—several exhibit strong

alignment with established DN pathways. For instance, imatinib targets platelet-derived growth factor receptor (PDGFR) tyrosine kinase signaling to blunt mesangial hyperproliferation (Yana Gvozdeva et al., 2025).

Across both Phase 1 whole-tissue screening and Phase 2 single-cell resolution, vorinostat emerged as the top-ranking therapeutic candidate. Vorinostat demonstrated exceptional performance across all datasets, maintaining high statistical confidence, strongly negative concordance scores, and target DEG reversal. Biologically, as an HDAC, vorinostat has been found to directly attenuate diabetes-associated renal hypertrophy, supppress pro-fibrotic signaling, and preserve glomerular architecture in diabetic models (Advani et al., 2011). Hence, its persistent identification across both whole-tissue T2DN screening and cell-type-resolved early DN profiling underscores its potential as a primary candidate for disease-modifying therapy.

Evaluation of Research Hypotheses

The experimental findings support the core hypotheses formulated in both phases of this multi-year investigation.

The Phase 1 hypothesis proposed that screening T2DN gene expression profiles through computational drug-matching platforms would identify small molecules capable of reversing disease-associated gene signatures and targeting key pathways involved in fibrosis and oxidative stress. This hypothesis was supported. Whole-tissue signature matching across PWB, L2S2, and CMap identified lead molecules—vorinostat, manumycin A, alvespimycin—that significantly reversed disease-associated transcriptional profiles ($p < 0.05$, $TIS \geq 90th$ percentile).

The Phase 2 hypothesis proposed that if cell-type-specific transcriptional signatures reveal DN mechanisms obscured in whole kidney tissue, then analyzing early DN through individual kidney cell types would identify distinct therapeutic compounds and biological processes compared to bulk-tissue analysis alone. This hypothesis was supported. Subtype-resolved snRNA-seq analysis uncovered unique cell-specific disease drivers—such as podocyte SASP activation—that were reduced within whole-tissue transcriptomes. Furthermore, single-cell profiling identified distinct lead therapeutics, including troglitazone, valproic acid, and calcitriol, that were entirely undetected in bulk-tissue screening alone.

Methodological Considerations

Certain biological and technical parameters constrain these findings. First, computational signature matching relies on cell line and biopsy data, which can't fully capture how cells actually interact inside a living human kidney. Second, because the single-cell analysis focused specifically on podocytes, endothelial cells, and mesangial cells, responses from other tubular cells weren't explicitly captured. Finally, while platforms like iLINCS and CMap cover thousands of molecules, a few structural analogs identified during virtual screening lacked full transcriptomic profiles in public databases, restricting my ability to test them computationally.

Future Directions

While this computational framework quickly narrowed down thousands of transcriptomic profiles into a focused list of leads, in vivo validation is the essential next step to confirm these predictions. Future studies should test the functional effects of top candidates—especially vorinostat, olaparib, and troglitazone—in living model systems. *Caenorhabditis elegans* (*C. elegans*) is an ideal starting point due to its conserved metabolic pathways, fast lifespan, and established use in metabolic disease research (Morcos & Hutter, 2009). And utilizng the genetic tools in *C. elegans*, this step will make it easier to directly verify whether these candidates modulate the exact target genes and pathways identified by the iLINCS and CMap pipelines.

Beyond wet-lab validation, updating the computational pipeline itself will strengthen these findings. For instance, in this study, only a select number of compounds were reanalyzed and updated using Reactome and DEG analysis. It would be helpful, therefore, to expand screening parameters beyond top-ranked candidates to capture additional high-priority targets that fell just below initial cutoff thresholds. Furthermore, applying this single-cell versus bulk framework to matched samples from the exact same patient cohorts (i.e. the same study) will eliminate cross-study variability and give a clearer picture of how cell-specific signals aggregate into whole-tissue disease states. Also, because this pipeline relies on open-source transcriptomic data, it can easily be adapted to uncover hidden subtype-specific targets across other complex multifactorial diseases.

Conclusion

The goal of this project was to develop a multi-scale, biologically driven drug discovery pipeline across two distinct phases of research to identify potential therapeutics for diabetic nephropathy. Phase 1 established a whole-tissue framework in T2DN to find compounds capable of reversing disease-associated gene expression, while Phase 2 expanded to single-cell resolution in early DN to compare subtype-specific signatures against bulk tissue profiles. As hypothesized, screening T2DN gene profiles in Phase 1 successfully prioritized small molecules targeting fibrosis and oxidative stress, and Phase 2 showed that analyzing individual kidney cell types reveals distinct candidate drugs and biological processes hidden in whole-tissue data. Podocyte-derived signals proved exceptionally robust and detectable at the whole-tissue level, indicating their central role in early disease. Among all candidates, the HDAC inhibitor vorinostat consistently reversed cell-type-specific signatures, targeting pro-fibrotic pathways while preserving filtration barrier integrity. Ultimately, this two-year study offers a practical computational framework to prioritize lead compounds for future in vivo testing. Beyond saving time and resources, finding viable small molecules could expand treatment options for patients and help shape a more efficient approach to computational drug discovery.

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